

Meshfree pricing and fast calibration of rough volatility: a physics-informed neural network for the Markovian-lifted PDE

[Authors / affiliations]

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Abstract

Rough-volatility models are non-Markovian and admit no finite-dimensional pricing partial differential equation (PDE); the Markovian lift restores one by approximating the fractional kernel with n exponential factors, at the cost of an $(n+1)$ -dimensional problem. We solve this lifted PDE with a physics-informed neural network (PINN) and ask how the network’s accuracy scales with the lifted dimension. Validating against an independent lifted Monte Carlo, and separating the network’s *solve-error* from the model’s *lift-error* in price space, we find the solve-error grows only sub-linearly—staying under about eleven price basis points to $n=32$ —while a naive finite-difference grid (50^{n+1} points) is already infeasible by $n=8$. The crossover between the two error sources gives a practical rule for choosing n . Calibrated to Deribit Bitcoin options the model returns a rough Hurst exponent ($H \approx 0.09$) with strong negative skew. A parametric PINN that takes the model parameters as inputs (with the spot variance and maturity fixed) is too coarse to calibrate on its own, but as a warm start for a short local optimization against the true pricer it recovers from-scratch fit quality at roughly nine times fewer true-model evaluations. The message is that where grid methods succumb to dimension, a mesh-free PINN degrades gracefully, and is most useful in practice as a differentiable warm start for calibration.

Keywords: physics-informed neural networks; rough volatility; Markovian lift; curse of dimensionality; option pricing; calibration.

1 Introduction

Pricing under a stochastic model is, underneath, a PDE, and for the standard models it is low-dimensional and easily gridded. Rough volatility breaks this: motivated by the empirical roughness of realized variance and the steep short-maturity skew (Gatheral et al., 2018; Bennedsen et al., 2022), the variance is driven by a fractional noise with Hurst exponent $H < 1/2$, which is non-Markovian and leaves no finite-dimensional pricing PDE (El Euch and Rosenbaum, 2019). The Markovian lift approximates the fractional kernel by a sum of n exponentials and recovers an $(n+1)$ -dimensional PDE (Abi Jaber and El Euch, 2019; Abi Jaber, 2019), trading intractability for high dimension—with the approximation order n sitting directly on the PDE dimension.

That coupling makes the lifted problem an ideal test for mesh-free neural PDE solvers. Physics-informed neural networks minimize the PDE residual at sampled points (Raissi et al., 2019) and, like other deep PDE solvers (Sirignano and Spiliopoulos, 2018; Han et al., 2018), avoid the exponential cost of a grid; they also learn from the equation itself (no labelled prices) and are differentiable

in their inputs, which we exploit for calibration. The closest prior work, Papapantoleon and Rou (2025), prices the lifted (rough) Heston PDE with a time-stepping deep gradient flow; we are not the first mesh-free solver here. Our contribution is a PINN residual formulation together with (i) a dimension-scaling result—the network’s solve-error grows sub-linearly while a naive grid explodes; (ii) a solve/lift error decomposition and a crossover rule for n ; (iii) calibration to real Bitcoin options; and (iv) a parametric-PINN warm-start that accelerates recalibration, reported honestly. We test everything on Bitcoin options, a liquid, rough, high-skew market.

2 Method

Under rough Heston the variance is driven by the kernel $K(t) = t^{\alpha-1}/\Gamma(\alpha)$, $\alpha = H + \frac{1}{2}$; the lift replaces it with $K(t) \approx \sum_i c_i e^{-x_i t}$, giving n Ornstein–Uhlenbeck factors U^i and $V = V_0 + \sum_i c_i U^i$. The discounted price $P(t, x, u)$, $x = \log S$, solves

$$0 = P_t + \left(r - \frac{1}{2}V\right)P_x + \frac{1}{2}VP_{xx} + \sum_i [-x_i u_i + \lambda(\theta - V)]P_{u_i} + \frac{1}{2}\nu^2 V \sum_{i,j} P_{u_i u_j} + \rho\nu V \sum_i P_{xu_i} - rP. \quad (1)$$

We solve (1) with a tanh PINN of (τ, x, u) , $\tau = T - t$, using the ansatz $P = P_{\text{base}}(\tau, x) + \tau N_\theta(\tau, x, u)$ that enforces the initial condition exactly, with P_{base} a Black–Scholes price anchored at the expected integrated variance. We add asymptotic boundary losses and, because PINNs are prone to optimization pathologies (Krishnapriyan et al., 2021; Wang et al., 2022), self-adaptive per-term loss weights (Kendall et al., 2018; McClenny and Braga-Neto, 2023) and a causal τ -curriculum (Wang et al., 2024); factor collocation is placed by a short pre-flight Monte Carlo.

We hold the network to two independent ground truths: a semi-analytic Fourier pricer (El Euch and Rosenbaum, 2019; Carr and Madan, 1999), exact for the model ($n \rightarrow \infty$), and a lifted Monte Carlo (an exponential-integrator Euler scheme), exact for the finite- n PDE the network actually solves. We therefore never report a single “PINN-versus-Fourier” number but decompose it, per strike and in price space where the signed errors add exactly,

$$\underbrace{(\text{PINN} - \text{Fourier})}_{\text{total}} = \underbrace{(\text{PINN} - \text{MC})}_{\text{solve-error}} + \underbrace{(\text{MC} - \text{Fourier})}_{\text{lift-error}}. \quad (2)$$

The solve-error is what training controls; the lift-error is a model approximation that only larger n can reduce. For calibration we compare differential evolution with the Fourier pricer, gradient descent on a parametric PINN (model parameters as inputs, V_0 and maturity fixed), and a hybrid that polishes the PINN’s guess with a few Fourier steps. Data are end-of-day BTC option quotes from Deribit (a cryptocurrency exchange and data provider); we set $r = 0$ and infer the forward from quotes.

3 Results

The network scales where the grid cannot. Training one architecture (width 64, depth 4) at $n = 4, 8, 16, 32$ on the canonical BTC calibration ($T = 0.15$) over three seeds, the solve-error against the lifted MC grows only mildly, from 5.4 ± 1.0 to 10.7 ± 2.0 price bp—about a doubling for an eight-fold rise in dimension (Figure 1). It is not flat (an early single-seed run misled us), but grows far more slowly than the dimension; with three seeds the adjacent- n bars overlap, so the robust statement is the two-fold growth over the full range. Training cost stays bounded (6–10 min, CPU). By contrast a naive full-tensor grid needs 50^{n+1} points— 3.1×10^8 , 2.0×10^{15} , 7.6×10^{28} ,

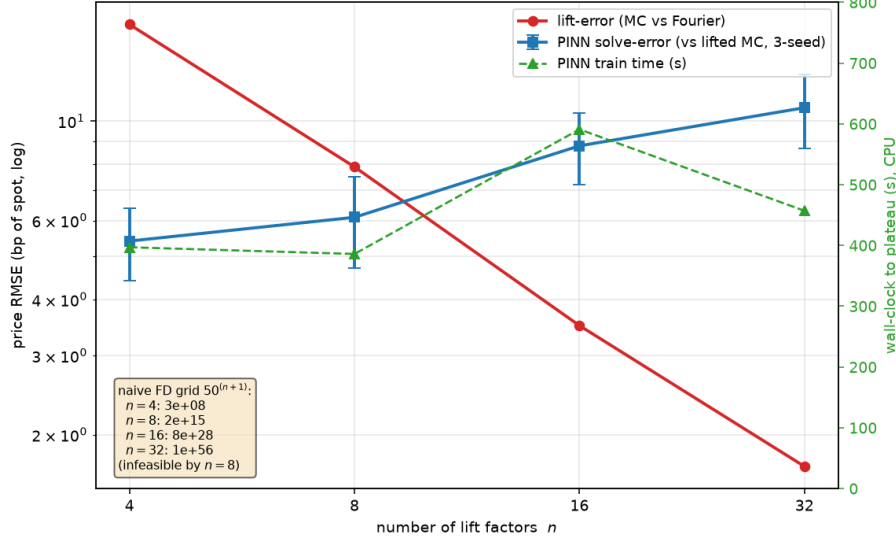


Figure 1: Dimension scaling. PINN solve-error (blue, 3-seed mean $\pm$ SD) grows sub-linearly and stays under ~ 11 price bp to $n = 32$, while the lift-error (red) falls and training time (green) stays bounded; the naive finite-difference grid (inset) is infeasible by $n = 8$.

1.2×10^{56} at $n = 4, 8, 16, 32$ —so even $n = 8$ (~ 16 PB) is infeasible. We are fair about this: 50^{n+1} is the *naive* grid, and sparse-grid/ADI/tensor-train schemes reach further but still degrade with n and are not standard for this PDE, so a competitive finite-difference solve at $n = 16$ – 32 is *impractical*, not provably impossible.

Decomposition and a rule for n . Split by (2), the historical “PINN vs Fourier” gap of 114–127 vol bp at $n = 4$ is almost entirely lift: 37 vol bp of solve-error on 133 of lift (partly cancelling to 121), and only 8 of solve-error with correlation switched off. The network already solves its PDE to well under 50 vol bp. Since the lift-error shrinks with n (133, 63, 27, 12 vol bp) while the solve-error climbs gently, they cross near $n = 8$ – 10 at $T \approx 0.15$: below it add factors, above it stop. The crossover moves with maturity—the lift-error grows with T (e.g. $8.6 \rightarrow 31.8$ price bp at $n = 4$ as $T : 0.05 \rightarrow 0.5$), so longer maturities need larger n .

Bitcoin is rough. A full-budget calibration to the Deribit BTC surface (differential evolution, 60 generations plus polish; 13 maturities, 97 points) returns $H = 0.090$, $\rho = -0.792$, $V_0 = 0.103$, $\theta = 0.253$, $\lambda = 2.567$, $\nu = 0.300$ at a vega-weighted RMSE of 496 vol bp. The Hurst exponent is firmly rough and the skew strongly negative. The fit is deliberately coarse: at 496 vol bp it reflects a known limitation of the lifted rough-Heston form on wide crypto smiles, a model-expressiveness issue orthogonal to the solve-error.

A parametric network and a warm-start hybrid. A single network over the five-parameter box costs accuracy: its solve-error at the box centre is 11.0 ± 4.7 price bp versus 5.9 ± 1.2 for a point-trained network at the same $n = 10$, and worse in the rough corners (up to 47.7). We therefore do not use it as a standalone calibrator. Calibrating the BTC $T \approx 0.15$ slice three ways (Table 1), Fourier-in-the-loop reaches 23.7 vol bp in 2018 Fourier evaluations (~ 168 s); the PINN alone is fast (~ 2 s, zero Fourier calls) but poor (222 ± 20 vol bp under the true model, partly because

Table 1: Calibration of the BTC $T \approx 0.15$ slice, three ways (3 PINN seeds; all judged by the true Fourier model over the same feasible region).

method	fit RMSE (vol bp)	wall-clock (s)	Fourier evals
Fourier-loop from scratch	23.7	168	2018
PINN-only (single-shot)	222 ± 20	~ 2	0
Hybrid (PINN + Fourier polish)	29.4 ± 7.1	23	237 ± 46

the slice optimum lies outside the training box). Seeding a short Fourier polish with the PINN’s guess reaches 29.4 ± 7.1 vol bp—on par with from-scratch on a 34–52% smile—in only 237 ± 46 Fourier evaluations (~ 23 s): about nine times fewer true-model calls and seven times faster. The parametric network is thus most useful as a differentiable warm start.

4 Conclusion

On the lifted rough-Heston PDE, a physics-informed neural network’s solve-error grows only sub-linearly in the lifted dimension while naive finite differences become infeasible—a concrete curse-of-dimensionality advantage on an economically real problem. Separating solve-error from lift-error is essential to read the numbers correctly and yields a crossover rule for the factor count. Applied to real Bitcoin options, the model is genuinely rough, and a parametric PINN, though not accurate enough to calibrate alone, is an effective warm start that recalibrates at roughly an order of magnitude fewer true-model evaluations. Natural next steps are adding the maturity as an input for whole-surface pricing, larger networks and GPUs to shrink the parametric penalty, and extensions to other rough models and path-dependent payoffs.

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